Mildred C. Templin. 1957. *Certain Language Skills in Children: Their Development and Interrelationships*, ned - new edition edition, volume 26. University of Minnesota Press.

Guy Tevet and Jonathan Berant. 2021. Evaluating the evaluation of diversity in natural language generation. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 326–346, Online. Association for Computational Linguistics.

Yufei Tian, Anjali Narayan-Chen, Shereen Oraby, Alessandra Cervone, Gunnar Sigurdsson, Chenyang Tao, Wenbo Zhao, Tagyoung Chung, Jing Huang, and Nanyun Peng. 2023. Unsupervised melody-to-lyrics generation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9235–9254, Toronto, Canada. Association for Computational Linguistics.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Adaku Uchendu, Jooyoung Lee, Hua Shen, Thai Le, Dongwon Lee, et al. 2023. Does human collaboration enhance the accuracy of identifying llm-generated deepfake texts? In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, volume 11, pages 163–174.

Ashwin Vijayakumar, Michael Cogswell, Ramprasaath Selvaraju, Qing Sun, Stefan Lee, David Crandall, and Dhruv Batra. 2018. Diverse beam search for improved description of complex scenes. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1).

Pablo Villalobos, Jaime Sevilla, Lennart Heim, Tamay Besiroglu, Marius Hobbhahn, and Anson Ho. 2022. Will we run out of data? an analysis of the limits of scaling datasets in machine learning. *arXiv preprint arXiv:2211.04325*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13484–13508, Toronto, Canada. Association for Computational Linguistics.

Alex Warstadt, Amanpreet Singh, and Samuel R. Bowman. 2019. Neural network acceptability judgments. *Transactions of the Association for Computational Linguistics*, 7:625–641.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837.

Chen Xing, Wei Wu, Yu Wu, Jie Liu, Yalou Huang, Ming Zhou, and Wei-Ying Ma. 2017. Topic aware neural response generation. In *Proceedings of the AAAI conference on artificial intelligence*, volume 31.

Canwen Xu, Daya Guo, Nan Duan, and Julian McAuley. 2023. Baize: An open-source chat model with parameter-efficient tuning on self-chat data. *arXiv preprint arXiv:2304.01196*.

Denis Yarats and Mike Lewis. 2018. Hierarchical text generation and planning for strategic dialogue. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 5591–5599. PMLR.

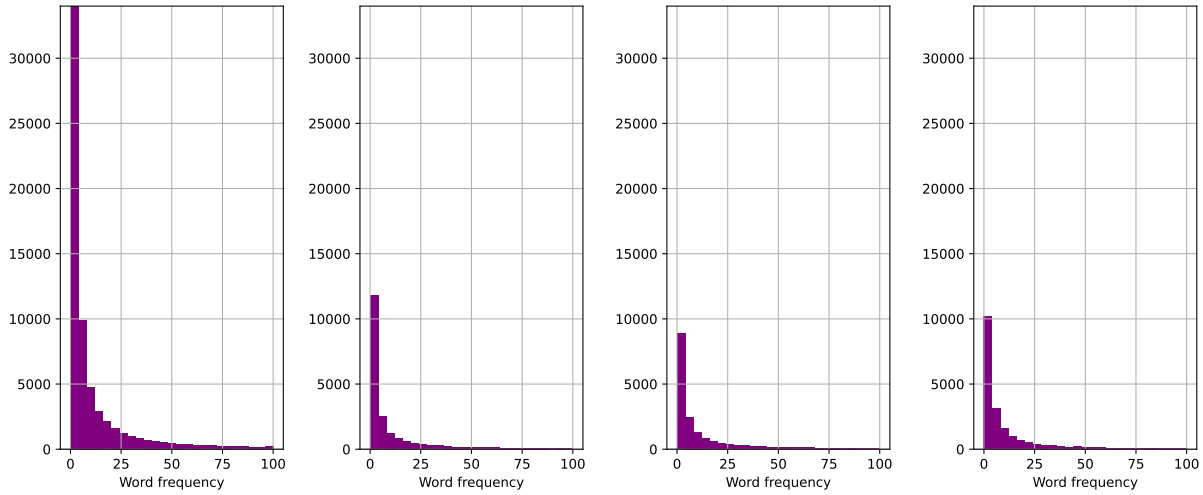
Hugh Zhang, Daniel Duckworth, Daphne Ippolito, and Arvind Neelakantan. 2021. Trading off diversity and quality in natural language generation. In *Proceedings of the Workshop on Human Evaluation of NLP Systems (HumEval)*, pages 25–33, Online. Association for Computational Linguistics.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. 2022. Opt: Open pre-trained transformer language models.

Weixiao Zhou, Gengyao Li, Xianfu Cheng, Xinnian Liang, Junnan Zhu, Feifei Zhai, and Zhoujun Li. 2023. Multi-stage pre-training enhanced by chatgpt for multi-scenario multi-domain dialogue summarization. *arXiv preprint arXiv:2310.10285*.

Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, and Yong Yu. 2018. Texygen: A benchmarking platform for text generation models. In *The 41st international ACM SIGIR conference on research & development in information retrieval*, pages 1097–1100.

# A Visualization of Diversity Metrics




(a) Human (b) Iteration 1 (c) Iteration 3 (d) Iteration 5

Figure 3: Histograms illustrating word frequency in texts produced across various iterations for the story generation task. For visual clarity, the x-axis, representing word frequency, is truncated at 100, though the actual distribution extends further. A noticeable trend is the diminishing presence of low-frequency, "unique" words in the synthetic text relative to human-generated text, a pattern that intensifies with each iteration. This trend highlights a progressive decline of lexical diversity in the generated text.

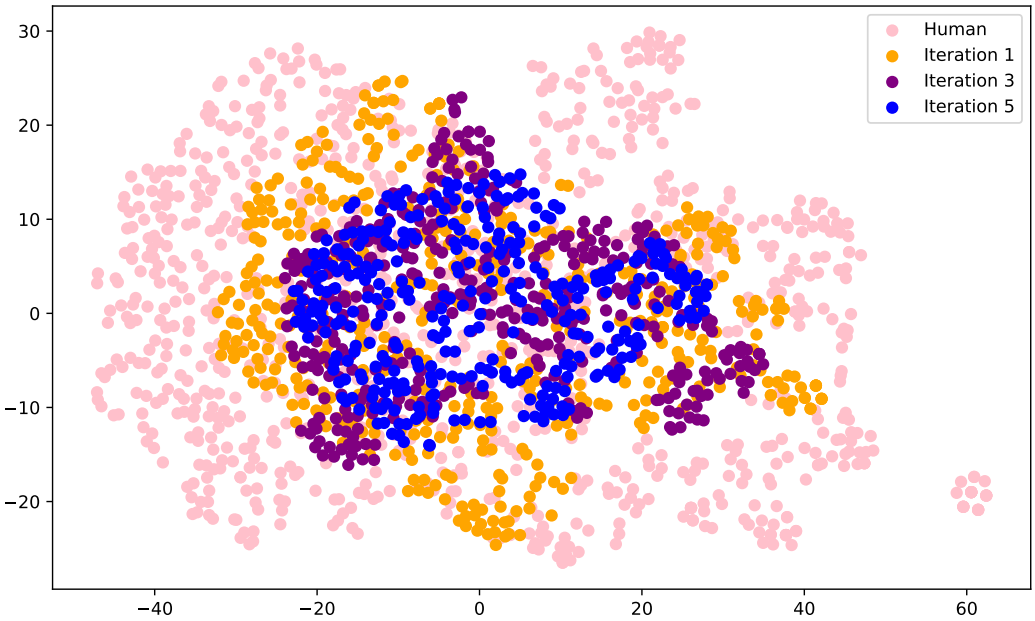



Figure 4: T-SNE visualization of dependency tree embeddings derived from sentences generated in successive iterations of our tuning-generation process. The visualization clearly depicts how, over time, the spatial distribution of the embeddings becomes increasingly compact. This decreasing spread is indicative of declining syntactic diversity.

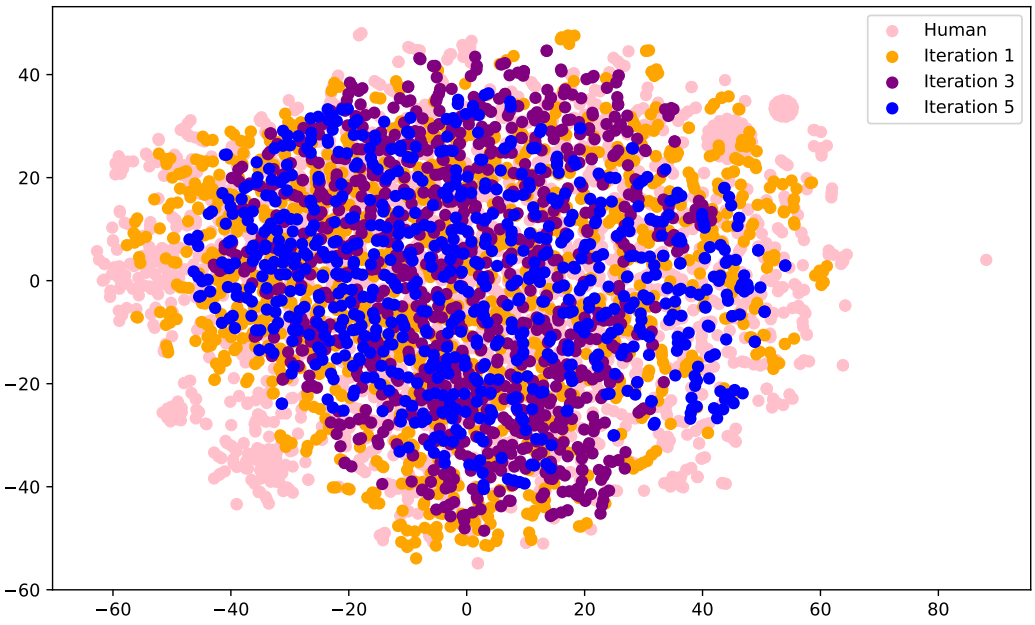



Figure 5: T-SNE visualization of sentence embeddings from text generated across different iterations. There is a noticeable decrease in dispersion over iterations, indicating a reduction in semantic diversity, though this change is less pronounced compared to that of syntactic diversity.

# B Implementation of Diversity Metrics

## B.1 Preprocessing

We apply preprocessing to the generated texts before computing the diversity metrics. For all three tasks, we remove the prompts from the generated texts. We remove <newline> tokens for story generation and replace URL links with WEBSITE for scientific abstract generation. We remove all punctuation marks for the calculation of lexical diversity metrics, but not for semantic diversity or syntactic diversity.

## B.2 Lexical Diversity Metrics

| **Iteration** | **Human** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| News Summarization | 18.89 | 18.21 | 18.42 | 18.75 | 19.01 | 19.25 | 19.71 |
| Scientific Abstract Generation | 49.35 | 48.80 | 49.61 | 49.60 | 49.64 | 49.60 | 49.50 |
| Story Generation | 148.77 | 149.77 | 149.92 | 149.92 | 149.91 | 149.96 | 149.84 |

Table 3: Average text lengths for post-truncation generations. Text lengths are measured by the number of words.

The distinct-$n$ metric varies as a function of text length, so we compute results at fixed lengths. We apply truncation to the generated texts, using different thresholds for each task: 20 for news summarization, 50 for scientific abstract generation and 150 for story generation. The average text lengths after truncation are presented in Table 3. We observe that all lengths consistently fall within a narrow range, allowing for fair comparison of distinct-$n$ results across iterations.

For Self-BLEU, we use a publicly available implementation[7]. We take the mean value of Self-BLEU-2 and Self-BLEU-3.

## B.3 Semantic Diversity Metrics

For sentence splitting, we use the NLTK sentence tokenizer. For Sentence-BERT, we use the all-mpnet-base-v2 model on huggingface. For the pairwise cosine distances, we randomly draw 2000 sentences for

[7] https://github.com/Danial-Alh/fast-bleu